HIM trimer form

Thursday, September 2, 2021 12:37 PM

Go to standard Au sample
Power on ET detector
Spot control = 4.0
Field of view SFIM mode = 1
Standby GFIS column

Mantainence -> GFIS -> Form the source

Form the source: maintain BIV source build, use active field -> begin
20min later it should be finished. Form completed: Succeeded on main screen

Process selection
Form the trimer
Select: ensure sample is under the column
-> begin: gas start
Column pressure ~e-6 torr
Also can control SFIM FOV in forming mode
SFIM controls: lens1 voltage fine&tracking
Imaging control: brightness 30%-50%
Dwell time 2us

Start increasing accelerator voltage =< extractor voltage
Stable accelerator to 30keV

L1 cannot exceed extractor voltage

1. Extractor -30kV
2. Accelerator 20kV
3. Len1 24kV

Image over the crossover of the atoms
=> related to accelerator and lens1

Adjust lens1 and contrast

Then only change extractor slowly to left to -35kV~-40kV, increase accelerator to 30keV
Extractor -39800V
When 3 atoms clearly seen in the FOV, reduce extractor

Done

Determine BIV
Begin
In graphics, draw a circle around the trimer
Right click -> measure (in graphics click M = measure) -> tune extractor to maximize the intensity
Maximum intensity extractor -32500V
Click Set BIV <- set the brightest extractor voltage as BIV

Next mechanically align the source

Graphics: circle one atom(brightest) above crossover and 1 under crossover
Tilt alignment to align the 2 circles

Mechanically shift the beam

Next set source distance: calibration to distance -104mm

If no trimer was found, check software first
System manager - service machine - hardware communication
Standby column, close Zen, close server, restart PC

Restart server, restart Zen, check hardware communications

Increase accelerator to 30keV, adjust lens 1 to much lower

Put finish when interrupted during forming new trimer

Unplug high voltage supply. Clean the connector with N2 and reconnect
Within 30min the trimer should be fine
Take a screenshot of the last stage